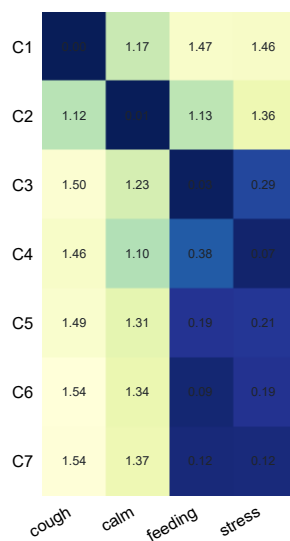


a

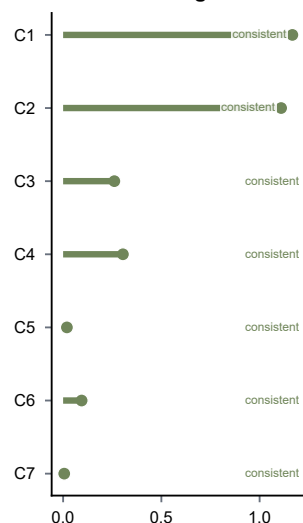
True class and complete Top-2 candidate evidence

Case	True class	Softmax Top-2	Main prototype Top-2	Subtype prototype Top-2
C1	cough	cough:0.999279 calm_grunt:0.000278	cough:1.000000 calm_grunt:0.000000	dry_cough:0.502413 abdominal_cough:0.497587
C2	calm_grunt	calm_grunt:0.999709 feeding:0.000143	calm_grunt:1.000000 cough:0.000000	calm_grunt:1.000000 abdominal_cough:0.000000
C3	feeding	feeding:0.971718 stress_vocal:0.027814	feeding:0.976441 stress_vocal:0.023559	feeding:0.974003 frightened_stress:0.023600
C4	stress_vocal	stress_vocal:0.980437 feeding:0.018823	stress_vocal:0.987218 feeding:0.012782	frightened_stress:0.985233 anxious_stress:0.008557
C5	feeding	feeding:0.582823 stress_vocal:0.413334	feeding:0.568187 stress_vocal:0.431813	anxious_stress:0.474453 feeding:0.473147
C6	stress_vocal	feeding:0.729500 stress_vocal:0.269693	feeding:0.793287 stress_vocal:0.206713	feeding:0.835862 frightened_stress:0.130336
C7	feeding	feeding:0.532451 stress_vocal:0.465917	feeding:0.519994 stress_vocal:0.480006	feeding:0.633301 frightened_stress:0.204193

b Main cosine distances



c Predicted margin and hierarchy consistency



d Hierarchical-route representative

